



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 2 Structured Questions

9729/02
12 September 2018
2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

For Examiner's Use	
1	/9
2	/9
3	/22
4	/13
5	/13
6	/9
TOTAL	/ 75

This document consists of **19** printed pages and **1** blank page.

1 Elements in Period 3 exhibit a variety of physical and chemical properties.

- (a)** An element in Period 3 has a high melting point but low electrical conductivity. Identify this element and explain why it has a high melting point using concepts of structure and bonding.

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[1]

- (b)** Two chlorides of Period 3 elements exist as liquids at room temperature. These two chlorides react with water to give white fumes, but only one of them gives a white solid.

- (i)** Using concepts of structure and bonding, explain why these two chlorides have low boiling points and exist as liquid at room temperature.

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[1]

- (ii)** Write chemical equations, with state symbols, for each of these chlorides reacting with water.

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[2]

(c) Period 3 elements also form oxides which reacts with water. Na_2O and SO_2 are two such oxides.

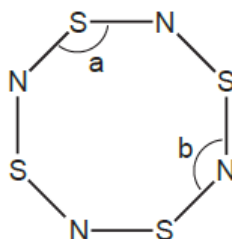
(i) Write chemical equations, with state symbols, when each of these oxides react with water.

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..... [2]

(ii) SO_2 is used as a food preservative. Suggest the property of SO_2 which enables it to function this way.

..... [1]

(d) Sulfur forms the compound S_4N_4 with nitrogen. The structure of S_4N_4 is shown below. Assume all bonds shown are single bonds.



Determine the number of lone pairs of electrons around a sulfur atom and a nitrogen atom in S_4N_4 .

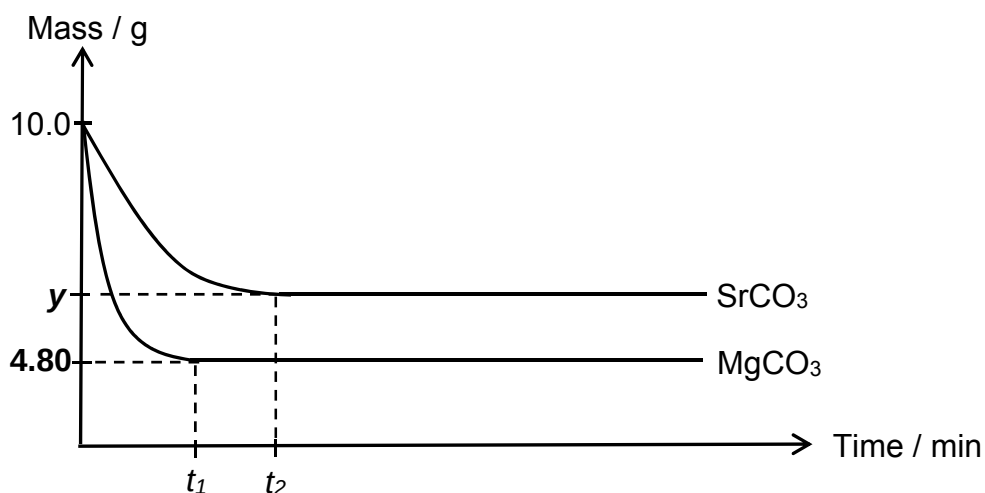
(i) Nitrogen atom: Sulfur atom: [1]

(ii) Which bond angle, **a** or **b**, will be smaller? Explain your answer.

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..... [1]

[Total: 9]

- 2 The following graph shows the change in mass with time for the decomposition of 10.0 g of magnesium carbonate and 10.0 g of strontium carbonate under the same conditions:



- (a) (i) Calculate the value of y .

[2]

- (ii) Explain why the value of t_2 is larger than t_1 .

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[3]

- (b) There are three bottles labelled **A**, **B** and **C** in the laboratory. Each bottle contains one of the following reagents: aqueous Cl_2 , KI solution and KBr solution.

The following tests were carried out and the results were summarised in the table below.

Experiment	Procedure	Observations
1	mixing reagent in bottle A with reagent in bottle B	mixture remains colourless
2	mixing reagent in bottle A with reagent in bottle C	mixture turns brown
3	mixing reagent in bottle B with reagent in bottle C	mixture turns brown

- (i) Which bottle contains aqueous Cl_2 ?
With the aid of a balanced equation, explain your answer.

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 [2]

- (ii) If hexane is also provided, how would you use it to identify the contents of the other two bottles? Include the observations in your answer.

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 [1]

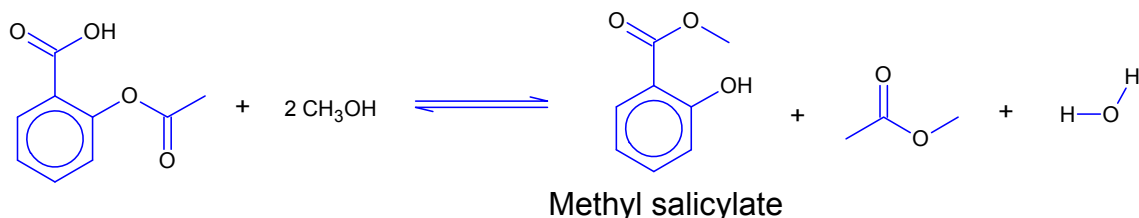
- (c) Using relevant data from the *Data Booklet*, comment on the thermal stability of hydrogen bromide and hydrogen chloride.

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 [1]

[Total:9]

- 3 Methyl salicylate, commonly known as oil of Wintergreen is used as a flavouring agent in candy. When methyl salicylate is applied to the skin, it causes a mild burning sensation which serves as a counter-irritant for sore muscles. It can be synthesise from aspirin. The reaction is as follows.



Reagent	Density / g cm ⁻³	M _r	Solubility in water
Acetylsalicylate acid (Aspirin)		180.0	insoluble
Sulfuric acid	1.84	98.0	soluble
Methanol	0.792	32.0	soluble
Methyl salicylate	1.17	152.1	insoluble
Water	1.00	18.0	

Preparation of impure methyl salicylate

1. Weigh approximately 20 g of acetylsalicylate acid into a round-bottom flask. Add 30 cm³ of methanol and stir until all the acetylsalicylate acid has dissolved.
2. Place 25 cm³ of concentrated sulfuric acid in the tap funnel and then add the acid dropwise into the reagents in the flask. Keep the contents well shaken and cool occasionally in an ice-water bath.
3. When all of the acid have been added, replace the tap funnel with a reflux condenser and gently boil the mixture for an hour.
4. Remove the condenser and boil off about 50% of the volume.

- (a) (i) By using the amounts given above, prove that methanol is in excess and suggest why it is done to increase yield.

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..... [2]

- (b) When concentrated sulfuric acid is added to the reaction mixture, cooling is necessary. Suggest the main cause of heat being produced at this stage.

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..... [1]

- (c) The reaction mixture was heated overnight. Why is this process necessary for the preparation of many covalent organic compounds?

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..... [1]

The crude product formed requires purification as it contains many impurities.

Purification of impure methyl salicylate

5. Transfer the remaining mixture into a separatory funnel. Shake the mixture with 30 cm³ of ice cool water and separate the aqueous layer from the methyl salicylate. Reject the aqueous layer.
6. Return the methyl salicylate to the funnel. Add 20 cm³ of dilute sodium hydrogencarbonate in the separating funnel, inverting the flask and opening the tap at intervals to prevent build up of pressure.
7. Transfer the methyl salicylate into a conical flask and add some granular anhydrous calcium sulfate. Swirl the mixture until the liquid is clear.
6. Filter the methyl salicylate into a clean vial. Weigh the purified product.

- (d) Volume of methyl salicylate produced is 10 cm³. Calculate the percentage yield.

[2]

- (e) The impure methyl salicylate was shaken with water (step 5) and the two layers are allowed to separate.

- (i) By considering the structure and bonding of methyl salicylate, suggest why it is an immiscible with methanol.

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[2]

- (ii) Suggest the organic compounds inside both layers. Explain your answer.

Top layer: Bottom layer:

Explanation:

[2]

- (f) (i) It is important to open the tap at intervals to prevent build up of pressure in when dilute aqueous sodium hydrogencarbonate was added (step 6). Explain, with the aid of an equation why this is important.

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[2]

- (g) (i) Draw the dot-and-cross diagram of calcium sulfate used in step 7.

[2]

- (ii) Calcium sulfate is a desiccant to remove any water left in the methyl salicylate product. When the desiccant has dissolved, the anions and cations are each surrounded by a number of water molecules.

Draw simple diagrams to show how a water molecule can be attached to a calcium cation, and to a sulfate anion. Label each of your diagram to show the type of interaction involved.

Ca^{2+} cation	SO_4^{2-} anion
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[2]

- (iii) The lattice energy of calcium sulfate is $-2640 \text{ kJ mol}^{-1}$.

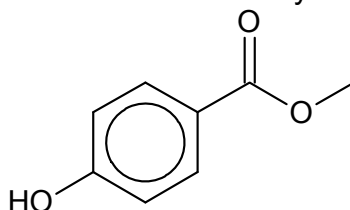
Use the following values of enthalpy change of hydration to construct an energy level diagram and use it to calculate the enthalpy change of solution, $\Delta H_{\text{sol}}^\ominus$, for calcium sulfate.

	$\Delta H_{\text{hyd}}^\ominus / \text{kJ mol}^{-1}$
Ca^{2+}	-1577
SO_4^{2-}	-1045



[3]

- (h) Methyl 4-hydroxybenzoate is an isomer of methyl salicylate.



methyl 4-hydroxybenzoate

Predict and explain whether methyl 4-hydroxybenzoate or methyl salicylate would have a higher boiling point. [3]

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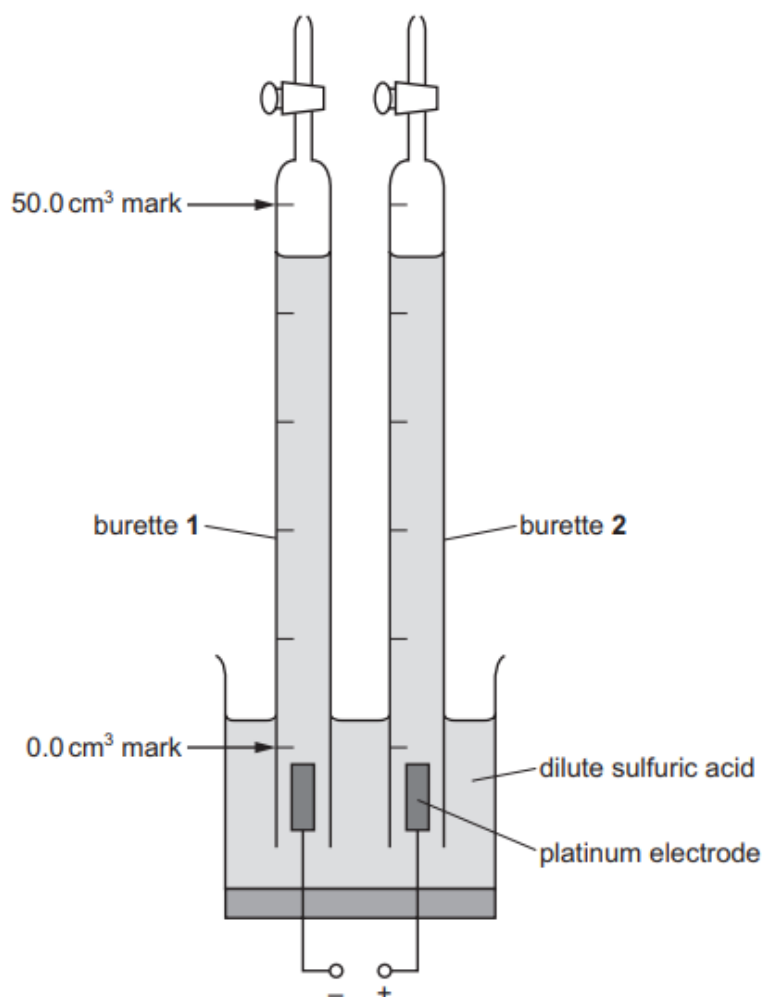
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[3]

[Total: 22]

- 4 Dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, can be electrolysed using platinum electrode and a direct current. Hydrogen gas is produced at the cathode and oxygen gas is formed at the anode. The two gases are collected separately in burettes filled with dilute sulfuric acid placed over each electrode.



Reaction at electrode in burette 1: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$

Reaction at electrode in burette 2: $\text{H}_2\text{O}(\text{l}) \rightarrow \frac{1}{2} \text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$

The production of hydrogen gas over time can be measured, and the data used to determine charge of one mole of electrons, known as the Faraday constant, F .

- (a) (i) Student A performed the experiment and the volume of hydrogen gas produced during the electrolysis process were recorded in the table.

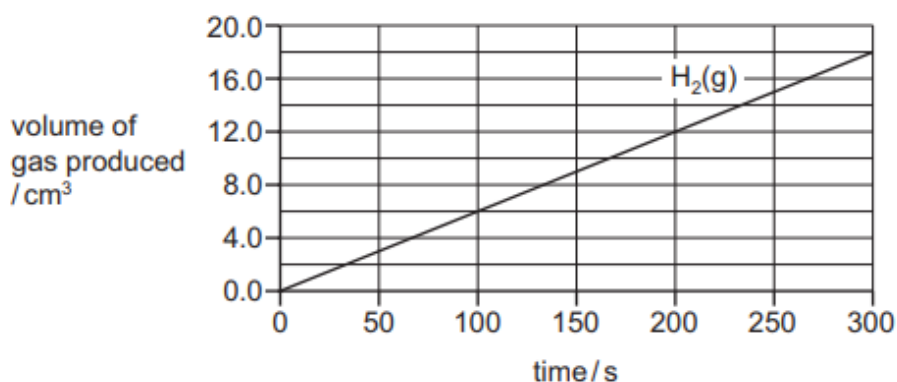
Process the results to calculate the volume of the hydrogen gas produced, in cm^3 , and the charge passed, in coulombs, C.

The current was kept constant at 0.80A.

Time/s	Reading on burette cm^3	Volume of hydrogen gas produced cm^3	Charge passed /C
0	46.20	0.00	
50	41.20		
100	36.20		
150	31.45		
200	25.80		
250	20.80		

[2]

- (ii) Another student B, performed the experiment at room temperature and his graph showing the relationship between volume of $\text{H}_2(\text{g})$ produced at the cathode and time was as shown below.



The gradient of the line of best fit gives the volume of hydrogen gas produced per coulomb.

Given that the gradient of the line to be $0.125 \text{ cm}^3 \text{ C}^{-1}$, calculate the number of moles of hydrogen gas produced per coulomb.

[Molar volume of gas = 24.0 dm^3 at room temperature and pressure]

[1]

- (iii) Using your answer from (a)(ii) and the half equation for the production of $\text{H}_2(\text{g})$, calculate a numerical value for the Faraday constant (the charge of 1 mole of electrons).

[1]

- (iv) Using the graph in (a)(ii), draw a line on the graph to show the relationship between volume of $\text{O}_2(\text{g})$ produced at the anode and time in this experiment.

[1]

- (v) Explain why the volume of $\text{O}_2(\text{g})$ measured in the experiment might be **less** than that shown by your drawn line. Assume that no gas is lost from leaks.

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[1]

- (b) A student suggested to replace the platinum electrodes with copper as it would be cheaper in the electrolysis of dilute sulfuric acid.

Quoting relevant data from the *Data Booklet*, suggest what effect, if any, the use of copper electrodes would have on the volume of gas produced at **each** electrode.

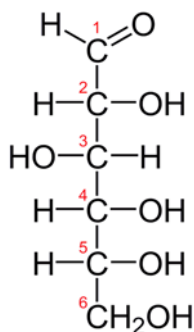
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[3]

- (c) Fehling's solution is a copper based chemical reagent used to differentiate between water soluble carbohydrate and ketone functional groups. This test was developed by German chemist Hermann von Fehling in 1849.



Fehling's solution can be used to screen glucose in urine, thus detecting diabetes.



Glucose

- (i) Suggest which carbon can be oxidised by Fehling's solution and draw the skeletal formula of the compound **D** which is the oxidised product of glucose after it has been acidified by dilute hydrochloric acid.

Carbon

Compound **D**:

[2]

- (ii) Determine the change in oxidation number on the carbon mentioned in (c)(i) with respect to glucose and compound **D**.

[1]

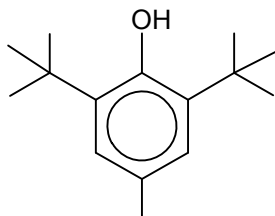
- (iii) Hot concentrated sulfuric acid was added to compound **D**. Compound **E** which consists of a 6-atoms ring structure was formed. Suggest the structure of compound **E**.

[1]

[Total: 13]

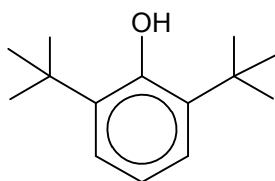
- 5 This question is about controversial food additives that are Generally Regarded As Safe (GRAS) by the American Food and Drug Administration (FDA), but are suspected to be harmful to humans when consumed.

(a) (i) Butylated hydroxytoluene (BHT) is useful for its antioxidant properties, to prevent oxidation in fluids (e.g. fuel, oil) and other foodstuffs where free radicals must be controlled.



Butylated hydroxytoluene (BHT)

Describe the mechanism for the synthesis of BHT using bromomethane and 2,6-di-tert-butylphenol shown below.



2,6-di-tert-butylphenol

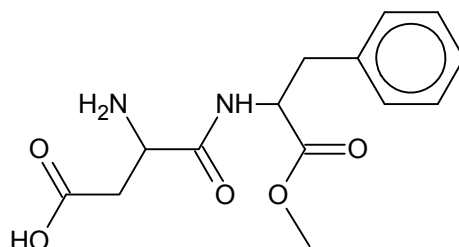
[3]

- (a) (ii) Propose a simple test-tube reaction to differentiate 2,6-di-tert-butylphenol and BHT. You are to clearly state the observation in your answer.

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[2]

- (b) Aspartame (APM) is an artificial non-carbohydrate sweetener used as a sugar substitute in some foods and beverages, which is especially useful for diabetic patients.



Aspartame (APM)

- (i) Label all chiral carbons with an (*) in the diagram above and calculate the total number of possible stereoisomers of APM.

[2]

- (ii) Draw all organic products formed when APM is reacted with a hot aqueous alkaline solution.

[3]

- (c) Potassium bromate, KBrO_3 , is typically used as a flour improver. It is a very strong oxidising agent, and allows for the baking of extremely white and fluffy bread, and it has the following half equation.



- (i) Suggest reagent and conditions for the oxidation of ethene to carbon dioxide.

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[1]

- (ii) Hot acidified bromate is also able to oxidise ethene into carbon dioxide. With reference to the *Data Booklet*, explain.

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[1]

- (iii) Hence, suggest how the yield of the carbon dioxide produced would be affected due to the use of bromate as an oxidising agent.

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[1]

[Total: 13]

- 6 Compound **G** and **H** have molecular formula of C_4H_8 and C_3H_6 respectively, and do not exhibit cis-trans isomerism.

Both **G** and **H** can decolourise hot potassium manganate(VII) to give compounds **J** and **K** respectively, together with a gas that produces a white precipitate when bubbled through calcium hydroxide.

J produces an orange precipitate when warm with 2,4–dinitrophenylhydrazine, and **K** produces effervescence when sodium hydrogencarbonate is added. **J** decolourises alkaline aqueous iodine but not **K**.

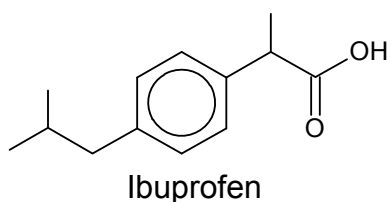
- (a) Draw the structures of **G**, **H**, **J**, and **K**.

[4]

- (b) Construct a balanced chemical equation on how **J** reacts with
- alkaline aqueous iodine,
 - 2,4-DNPH.

[2]

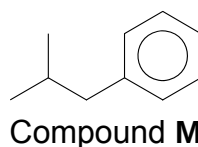
- (c) Ibuprofen is an oral or intravenous medication that is used for treating pain, fever, and inflammation. It typically begins working within an hour and it has the following structure.



Compounds **G** and **H** from (a) are both prerequisites for the synthesis of Ibuprofen.

The reaction schema to obtain ibuprofen is described below.

- (1) HBr gas is introduced to **G** to form **L**.
- (2) **L** is reacted with benzene in the presence of anhydrous FeBr₃. Compound **M** is obtained.



- (3) Aqueous Br₂ is added to **H** to form **N**.
- (4) **N** and **M** are reacted in the presence of anhydrous FeBr₃. Compound **P** with molecular formula C₁₃H₂₀O is produced.
- (5) Hot acidified potassium dichromate(VI) is then added to **P** to produce ibuprofen.

Using the structure of ibuprofen and the information provided, draw the structures of **L**, **N** and **P**.

[3]

[Total: 9]

END OF PAPER

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